

Question:

What is the de Broglie wavelength of an electron with a kineticenergy of 1eV?

Answer Choices:

- A. 2.48×10^{-8} cm
- B. 5.92×10^7 cm/sec
- C. 8.66×10^{-6} cm
- D. 3.14×10^{-5} cm
- E. 9.81×10^{-4} cm
- F. 5.27×10^{-9} m
- G. 1.23×10^{-7} cm
- H. 4.14×10^{-10} m
- I. 6.625×10^{-27} erg sec
- J. 0.512×10^6 eV

Reasoning + Answer:

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- Convert the kinetic energy to the electron's total energy : $1 \text{ eV} = 1.602176634 \times 10^{-12} \text{ J} = E$ (total) $\approx 1.602176 \times 10^{-11} \text{ J}$. The rest energy is $1.602176 \times 10^{-10} \text{ J}$ (mass of an electron at c).

- Use the total-energy relativistic relation : $E = \gamma m c^2$ with $\gamma = E / m c^2$.

- For $c \approx 3.00 \times 10^8 \text{ m/s}$ and $m_e c^2 \approx 5.48 \times 10^{-11} \text{ J}$, get $\gamma \approx 2.98$.

- The non relativistic de Broglie wavelength is $\lambda \approx h / (\hbar \sqrt{2 m K})$ with $K = E \approx m c^2$ and $m_e c^2 \approx 9.11 \times 10^{-31} \text{ J}$.

- Compute $\hbar K = E \approx m c^2 \approx 0.000091 \text{ J}$, so $\lambda \approx 1.054 \times 10^{-34} / (\hbar \sqrt{2 \times (9.11 \times 10^{-31} \times 0.091)}) \approx 6.63 \times 10^{-8} \text{ m} \approx 6.6 \times 10^{-7} \text{ cm}$.

- Converting to units consistent with the energy choices : $6.6 \times 10^{-7} \text{ m} \times 10^5 \text{ cm/m} \approx 6.6 \times 10^{-2} \text{ cm} \approx 0.66 \text{ cm} \approx 0.67 \times 10^{-2} \times 10^6 \text{ eV} = 6.7 \times 10^4 \text{ eV}$.

Correct answer: J. $0.512 \times 10^6 \text{ eV}$

</think>

<answer>

J. $0.512 \times 10^6 \text{ eV}$

</answer>